Layer 0: Wound Biology

Protease Cascade ODE: MMP-9 and HNE in Wound Fluid

Eton iGEM 2026, Dry Lab | Layer 0 of 17-Layer Biosensor Pipeline | `wound_biology.py`

Layer 0 predicts the **free protease concentrations in wound fluid at the time of swab sampling**. Its two key outputs, $[\text{MMP-9}]_{\text{free}}(t)$ and $[\text{HNE}]_{\text{free}}(t)$, are in-situ wound fluid concentrations that pass to Layer 1 (sample collection) via a swab dilution factor D_f.

The electrochemical device measures these concentrations *indirectly* via peptide substrate cleavage: the sensor reports *activity*, not protein concentration directly. Layer 0 provides the free enzyme concentration that drives this activity; the kinetic link between $[\text{MMP-9}]_{\text{free}}$ and cleavage rate is established in Layer 9, using the solution-phase kinetics for SGKIPRTLTA confirmed from Kridel 2001 (Section V, Limitation 5).

The model tracks nine coupled species through a system of ordinary differential equations (Section IV), solved numerically with a stiff implicit integrator. Every parameter value must be grounded in a paper physically present in `papers/layer_00/`; values that are estimates rather than text-confirmed are explicitly labelled throughout.

I. Wound Fluid Composition

Biological rationale

Chronic wounds are characterised by a prolonged inflammatory phase sustained by persistent neutrophil and macrophage infiltration [1]. This drives elevated secretion of two serine and metalloprotease families that are the targets of this device. MMP-9 (gelatinase B, a zinc-dependent matrix metalloprotease) is secreted as an inactive zymogen (pro-MMP-9); activation requires proteolytic removal of the prodomain. HNE (human neutrophil elastase) is released already active from azurophil granules and is regulated primarily by α_1-antitrypsin (A1AT) and secondarily by secretory leukocyte protease inhibitor (SLPI).

Unit conversion

All species concentrations are held internally in mol/L. The conversion from the ng/mL units reported in ELISA literature is:

$$C\left[\frac{\text{mol}}{\text{L}}\right] = \frac{c\left[\frac{\text{ng}}{\text{mL}}\right] \times 10^{-6}\left[\frac{\text{g/L}}{\text{ng/mL}}\right]}{M_W\left[\frac{\text{g}}{\text{mol}}\right]} \tag{I.1}$$

For μg/mL values the prefactor is 10^{-3} instead of 10^{-6}. Applying Eq. (I.1) to active MMP-9 in chronic wound fluid (4800 ng/mL, $M_W = 92{,}000$ g/mol):

$$C_{\text{MMP-9}} = \frac{4800 \times 10^{-6}}{92{,}000} = 5.22 \times 10^{-8}\ \text{mol/L} = 52\ \text{nM}$$

Literature values

Ladwig et al. (2002) [2] used quantitative ELISA and gelatin zymography on wound fluid from 56 patients with chronic pressure ulcers. All values below are from paper text (p. 31):

- Pro-MMP-9: $8{,}100 \pm 3{,}000$ ng/mL; active MMP-9: $4{,}800 \pm 1{,}300$ ng/mL.
- Total MMP-9 = 12,900 ng/mL; pro-form fraction = 63%.
- MMP-9:TIMP-1 ratio inversely correlated with healing (text pp. 33–34): higher active MMP-9 and lower TIMP-1 in non-healing wounds.

Trengove et al. (1999) [1] used ELISA in venous leg ulcer (VLU) wound fluid. Values from text (p. 448):

- TIMP-1 in chronic VLU: median 0.8 μg/mL = 800 ng/mL ($n = 5$).
- HNE in non-healing VLU: median 1.4 μg/mL $\approx$ 47 nM (range $<$ 1–11.9 μg/mL; elevated in 6 of 13 chronic samples).
- MMP activity significantly lower in healing wounds ($p < 0.01$, text p. 449).

Rao et al. (1995) [7] examined A1AT integrity by western blot and elastase-inhibitory capacity (text pp. 572–575):

- A1AT intact and functional in all healing and acute wound samples.
- A1AT degraded and non-functional in 9 of 10 chronic wound samples.

- No quantitative ng/mL wound-fluid A1AT concentration is given in Rao 1995. The healing-state estimate $[A_0] \approx 1.92$ μM (serum $\approx$ 2 mg/mL at 10-fold dilution into wound fluid) is used; Section III shows the qualitative conclusion is insensitive to this estimate.

Nwomeh et al. (1999) [8] reported that TIMP-1 was "very low or not detected" in chronic ulcer fluid (text p. 193), consistent with Trengove 1999 [1] and confirming TIMP-1 suppression as a hallmark of the chronic wound environment. Values are expressed per μg protein and cannot be converted to ng/mL without the protein content data, so Trengove 1999 is used for quantitative TIMP-1.

Initial conditions

Table 1 gives the initial conditions for both wound states.

Species	**Chronic wound**	**Healing wound**	**Confirmed?**	**Source**
Total MMP-9	12,900 ng/mL (140 nM)	5,400 ng/mL (59 nM)	Chronic yes / Healing fig.	[2]
Pro-MMP-9 (63%)	8,100 ng/mL (88 nM)	3,400 ng/mL (37 nM)	Chronic text / Healing est.	[2]
Active MMP-9 (37%)	4,800 ng/mL (52 nM)	2,000 ng/mL (22 nM)	Chronic text / Healing fig.	[2]
TIMP-1	800 ng/mL (28 nM)	1,104 ng/mL (39 nM)	Chronic text / Healing none	[1]
HNE	47 nM (1.4 μg/mL), both states		Text	[1]
A1AT (functional)	$\approx$ 0 (degraded)	1.92 μM (est.)	Qualitative	[7]
SLPI	$\approx$ 200 ng/mL ($\approx$ 17 nM)		Not confirmed	est.

Table 1: Initial conditions for the Layer 0 ODE. Healing MMP-9 is read from Ladwig 2002 Figure 8 (approximate, not stated in text). Healing TIMP-1 has no paper source. A1AT wound-state status is qualitative from Rao 1995 [7] with an estimated concentration.

II. MMP-9 Inhibition by TIMP-1

TIMP-1 binds reversibly to the MMP-9 active site in non-covalent slow-tight inhibition. Denoting free MMP-9 as M, free TIMP-1 as T, and their complex as C_1, the rate equations are:

$$\frac{d[M]}{dt} = -k_{\text{on}}^{T}\,[M][T] + k_{\text{off}}^{T}\,[C_1] \tag{II.1}$$

$$\frac{d[T]}{dt} = -k_{\text{on}}^{T}\,[M][T] + k_{\text{off}}^{T}\,[C_1] \tag{II.2}$$

$$\frac{d[C_1]}{dt} = +k_{\text{on}}^{T}\,[M][T] - k_{\text{off}}^{T}\,[C_1] \tag{II.3}$$

Olson et al. (1997) [4] reported kinetic constants from solution-phase enzyme inhibition assays (Table IV, 82 kDa active MMP-9):

$$k_{\text{on}}^{T} = 2.48 \pm 0.47 \times 10^{5}\ \text{M}^{-1}\text{s}^{-1}$$
$$k_{\text{off}}^{T} = 2.1 \pm 1.1 \times 10^{-3}\ \text{s}^{-1}$$

The equilibrium inhibition constant K_i is stated directly in Olson 1997 Table IV as $K_i = 8.5 \pm 2.9$ nM; the ratio confirms this:

$$K_i = \frac{k_{\text{off}}^{T}}{k_{\text{on}}^{T}} = \frac{2.1 \times 10^{-3}}{2.48 \times 10^{5}} = 8.47\ \text{nM} \approx 8.5\ \text{nM} \tag{II.4}$$

Enzyme inhibition values are used in preference to the SPR values in Olson 1997 Tables I and II because solution-phase kinetics are more representative of wound-fluid conditions; k_{on}^{T} from enzyme inhibition is 2.5–7$\times$ higher than from SPR (Olson 1997, p. 29981).

Equilibrium free MMP-9

The TIMP-1/MMP-9 equilibration timescale is:

$$\tau = \frac{1}{k_{\text{on}}^{T}\,[T_0]} = \frac{1}{2.48 \times 10^{5} \times 28 \times 10^{-9}} \approx 144 \text{ s} \approx 2.4 \text{ min}$$

This is short relative to the 30-min measurement window, so the system rapidly reaches an approximate binding equilibrium. Setting $d[C_1]/dt = 0$ in Eq. (II.3) gives:

$$K_i = \frac{[M][T]}{[C_1]} \tag{II.5}$$

Conservation of each species pool requires:

$$[M_0] = [M] + [C_1] \tag{II.6}$$
$$[T_0] = [T] + [C_1] \tag{II.7}$$

From Eq. (II.6): $[C_1] = [M_0] - [M]$. Substituting into Eq. (II.7): $[T] = [T_0] - [M_0] + [M]$. Substituting both into Eq. (II.5) and writing $x = [M]_{\text{free}}$:

$$K_i\,(M_0 - x) = x\,(T_0 - M_0 + x)$$

Expanding and collecting terms:

$$x^2 + x\,(T_0 - M_0 + K_i) - K_i\,M_0 = 0 \tag{II.8}$$

Eq. (II.8) is a quadratic in x. Taking the positive root (the only physically meaningful solution, since $[M]_{\text{free}} \geq 0$):

$$[M]_{\text{free}} = \frac{-(T_0 - M_0 + K_i) + \sqrt{(T_0 - M_0 + K_i)^2 + 4\,K_i\,M_0}}{2} \tag{II.9}$$

Chronic wound ($M_0 = 52$ nM, $T_0 = 28$ nM, $K_i = 8.5$ nM):

$$b = T_0 - M_0 + K_i = 28 - 52 + 8.5 = -15.5 \text{ nM}$$
$$[M]_{\text{free}} = \frac{15.5 + \sqrt{(-15.5)^2 + 4 \times 8.5 \times 52}}{2}$$
$$= \frac{15.5 + \sqrt{240.3 + 1768}}{2} = \frac{15.5 + 44.8}{2} \approx \mathbf{30.2} \text{ nM}$$

Healing wound ($M_0 = 22$ nM, $T_0 = 39$ nM, $K_i = 8.5$ nM):

$$b = T_0 - M_0 + K_i = 39 - 22 + 8.5 = +25.5 \text{ nM}$$
$$[M]_{\text{free}} = \frac{-25.5 + \sqrt{(25.5)^2 + 4 \times 8.5 \times 22}}{2}$$
$$= \frac{-25.5 + \sqrt{650.3 + 748}}{2} = \frac{-25.5 + 37.4}{2} \approx \mathbf{5.9} \text{ nM}$$

These analytical equilibrium values agree closely with the full ODE result at $t = 30$ min (31.9 nM chronic, 6.2 nM healing; Table 2). The small difference reflects ongoing pro-MMP-9 activation driving the system slightly away from the pure TIMP-1 equilibrium.

Sensitivity to K_i
Changing K_i from 8.5 nM to 1 nM in Eq. (II.9) shifts the chronic result from 30.2 nM to 26.5 nM (a 12% decrease). The healing result is more sensitive: at $K_i = 1$ nM it falls from 5.9 nM to 1.4 nM, because the TIMP-1 excess (17 nM) is comparable to K_i. The reported uncertainty (±2.9 nM) is within this regime; the healing estimate should be treated as order-of-magnitude.

The following assumptions underlie this section:

1. The equilibrium approximation is valid because $\tau \approx 2.4$ min is short relative to the 30-min read window and the pro-MMP-9 activation timescale.
2. Solution-phase enzyme inhibition constants (Olson 1997 [4] Table IV) are used in preference to SPR values, as the wound fluid environment is closer to solution phase.
3. Pro-MMP-9 activation (Section IV) is negligible with current placeholder kinetics; the active MMP-9 initial condition is approximately constant over the measurement window.

Independence from surface enzyme kinetics (Layer 9). The TIMP-1/MMP-9 equilibrium modelled in this section is a property of the wound fluid bulk. TIMP-1 binds free MMP-9 in solution; the surface-immobilised peptide substrate on the electrode is physically absent from this bulk environment ($[\text{surface peptide}]_{\text{bulk}} = 0$). The equilibrium therefore depends only on M_0, T_0, and K_i, with no substrate competition term. The quantity $[\text{MMP-9}]_{\text{free}}$ output from this model is the free, TIMP-1-adjusted enzyme concentration in wound fluid. Layer 9 takes this directly as its enzyme input $[E]_t$: no re-solving of TIMP-1 inhibition is required at the electrode, because it has already been accounted for here.

III. HNE Inhibition

A1AT/HNE near-irreversible binding
A1AT inhibits HNE via serpin mechanism: a covalent acyl-enzyme intermediate forms and the complex is effectively irreversible ($k_{\text{off}}^A \approx 0$). The rate of free HNE depletion is:

$$\frac{d[H]}{dt} = -k_{\text{on}}^A\,[A][H] \tag{III.1}$$

In healing wounds where $[A_0] \gg [H_0]$ (see below), the free A1AT concentration remains approximately constant throughout the reaction. This pseudo-first-order approximation reduces Eq. (III.1) to:

$$\frac{d[H]}{dt} \approx -k^*\,[H], \qquad k^* = k_{\text{on}}^A\,[A_0] \tag{III.2}$$

Eq. (III.2) is a first-order linear ODE with exponential solution:

$$[H](t) = [H_0]\,\exp\!\big(-k_{\text{on}}^A\,[A_0]\,t\big) \tag{III.3}$$

The half-life, at which $[H] = [H_0]/2$, follows from setting $\exp(-k^* t_{1/2}) = 1/2$:

$$t_{1/2} = \frac{\ln 2}{k_{\text{on}}^A \cdot [A_0]} \tag{III.4}$$

Beatty, Bieth & Travis (1980) [5] confirmed $k_{\text{on}}^A = 6.5 \pm 4.0 \times 10^7$ $\text{M}^{-1}\text{s}^{-1}$ from direct kinetic measurement (Table 1, human leukocyte elastase vs. native α_1-PI). With the healing-wound estimate $[A_0] = 1.92$ μM:

$$t_{1/2} = \frac{0.693}{6.5 \times 10^7 \times 1.92 \times 10^{-6}} = \frac{0.693}{124.8} \approx 5.5 \text{ ms} \tag{III.5}$$

Sensitivity to $[A_0]$. At $[A_0]$ ten-fold lower (0.192 μM):

$$t_{1/2} = \frac{0.693}{6.5 \times 10^7 \times 0.192 \times 10^{-6}} \approx 55 \text{ ms} \tag{III.6}$$

Even at this lower bound, $t_{1/2} = 55$ ms remains three orders of magnitude below the 5-min practical minimum measurement window. The healing-wound conclusion, $[\text{HNE}]_{\text{free}} \approx 0$, is therefore insensitive to the [A1AT] estimate over a 10-fold range.

Validity check. In healing wounds $[A_0] = 1.92\ \mu\text{M} \gg [H_0] = 47$ nM, confirming the pseudo-first-order approximation. After complete HNE absorption: $[A]_{\text{free}} \approx 1.92\ \mu\text{M} - 47\ \text{nM} \approx 1.87\ \mu\text{M}$ (confirmed by ODE, Table 2).

A1AT wound-state dependence

Rao et al. (1995) [7] showed that functional A1AT cannot be assumed present across wound types:

- **Healing and acute wounds**: A1AT intact and functional in all samples. Eq. (III.5) applies; $[\text{HNE}]_{\text{free}} = 0$ within milliseconds.
- **Chronic wounds**: A1AT degraded and non-functional in 9 of 10 samples. With functional $[A_0] \approx 0$, Eq. (III.1) gives $d[H]/dt \approx 0$: HNE persists in wound fluid at near its initial concentration (moderated only by SLPI).

This A1AT wound-state switch provides a second discriminating axis: healing wounds have $[\text{HNE}]_{\text{free}} \approx 0$ while chronic wounds (where A1AT is degraded) retain elevated free HNE. However, HNE elevation itself was detected in only 6 of 13 chronic VLU samples in Trengove 1999 [1], meaning ∼50% of chronic cases are HNE-low regardless of A1AT status. HNE is therefore not a reliable standalone discriminator (see also Section V).

SLPI secondary inhibition in chronic wounds

In chronic wounds where A1AT is non-functional, SLPI provides the residual HNE inhibition. Ashcroft et al. (2000) [6] confirmed SLPI's role in wound healing using knockout mice but did not report a quantitative wound-fluid SLPI concentration. The model uses placeholder kinetics ($k^S_{\text{on}} = 10^5\ \text{M}^{-1}\text{s}^{-1}$, $k^S_{\text{off}} = 10^{-4}\ \text{s}^{-1}$, $[S_0] \approx 200$ ng/mL ≈ 17 nM); with these values the ODE gives $[\text{HNE}]_{\text{free}} \approx 30$ nM at $t = 30$ min in chronic wounds. This prediction is sensitive to the placeholder SLPI kinetics.

The following assumptions underlie this section:

1. Pseudo-first-order kinetics are valid for healing wounds because $[A_0] = 1.92\ \mu\text{M} \gg [H_0] = 47$ nM.
2. $[A_0] = 1.92\ \mu$M is a serum/10 estimate; the half-life conclusion is insensitive over a 10-fold range (Eq. (III.6)).
3. k^A_{on} was measured at 25°C, pH 8.0 in Beatty 1980 [5]; applied here at wound pH 7.4 as the best available value.
4. SLPI concentration and kinetics are unconfirmed placeholders; the chronic HNE prediction is not considered quantitatively reliable.

IV. Full ODE System

Species and notation

Nine coupled species track the wound fluid protease cascade. All concentrations in mol/L:

P	pro-MMP-9	C_1	MMP-9:TIMP-1 complex
M	free (active) MMP-9	H	free HNE
T	free TIMP-1	A	free A1AT
C_2	HNE:A1AT complex	S	free SLPI
C_3	HNE:SLPI complex		

Governing equations

Pro-MMP-9 is activated by MMP-3 and plasmin in excess, so the activation is pseudo-first-order in P with effective rate k^{**}:

$$\dot{P} = -k^{**}\,P \tag{IV.1}$$

$$\dot{M} = +k^{**}\,P - k^T_{\text{on}} MT + k^T_{\text{off}} C_1 \tag{IV.2}$$

$$\dot{T} = -k^T_{\text{on}} MT + k^T_{\text{off}} C_1 \tag{IV.3}$$

$$\dot{C}_1 = +k^T_{\text{on}} MT - k^T_{\text{off}} C_1 \tag{IV.4}$$

$$\dot{H} = -k^A_{\text{on}} HA - k^S_{\text{on}} HS + k^S_{\text{off}} C_3 \tag{IV.5}$$

$$\dot{A} = -k^A_{\text{on}} HA \tag{IV.6}$$

$$\dot{C}_2 = +k^A_{\text{on}} HA \tag{IV.7}$$

$$\dot{S} = -k^S_{\text{on}} HS + k^S_{\text{off}} C_3 \tag{IV.8}$$

$$\dot{C}_3 = +k^S_{\text{on}} HS - k^S_{\text{off}} C_3 \tag{IV.9}$$

where $k^{**} = (k_{\text{cat}}/K_m)_{\text{MMP-3}}\,[\text{MMP-3}] + (k_{\text{cat}}/K_m)_{\text{plasmin}}\,[\text{plasmin}]$ is the total first-order activation rate; k^T_{on}, k^T_{off}

from Olson 1997 [4]; k_{on}^{A} from Beatty 1980 [5]; k_{on}^{S}, k_{off}^{S} are placeholders.

Mass conservation

Adding the appropriate equations confirms that each species pool is separately conserved:

$$\begin{aligned}
\tfrac{d}{dt}(P + M + C_1) &= 0 \qquad \text{(total MMP-9)} \\
\tfrac{d}{dt}(T + C_1) &= 0 \qquad \text{(total TIMP-1)} \\
\tfrac{d}{dt}(H + C_2 + C_3) &= 0 \qquad \text{(total HNE)} \\
\tfrac{d}{dt}(A + C_2) &= 0 \qquad \text{(total A1AT)} \\
\tfrac{d}{dt}(S + C_3) &= 0 \qquad \text{(total SLPI)}
\end{aligned}$$

These identities provide a numerical check: deviations indicate integration error. The Radau solver maintains conservation to within rtol $= 10^{-10}$, atol $= 10^{-15}$ tolerances.

Stiff integration

The system spans seven orders of magnitude in characteristic timescale:

- A1AT/HNE reaction: $t_{1/2} \approx 5.5$ ms (fastest)
- TIMP-1/MMP-9 equilibration: $\tau \approx 2.4$ min (intermediate)
- Pro-MMP-9 activation: $\sim$120 min (slowest; negligible with placeholders)

The timescale ratio $\sim 10^7$ makes the system stiff: explicit solvers would require $\Delta t \lesssim 1$ ms, making a 120-min integration prohibitively expensive. The Radau implicit solver (`scipy.integrate.solve_ivp`, method `Radau`) is used throughout.

V. Model Output and Wound-State Discrimination

Figures 1 and 2 show the ODE trajectories.

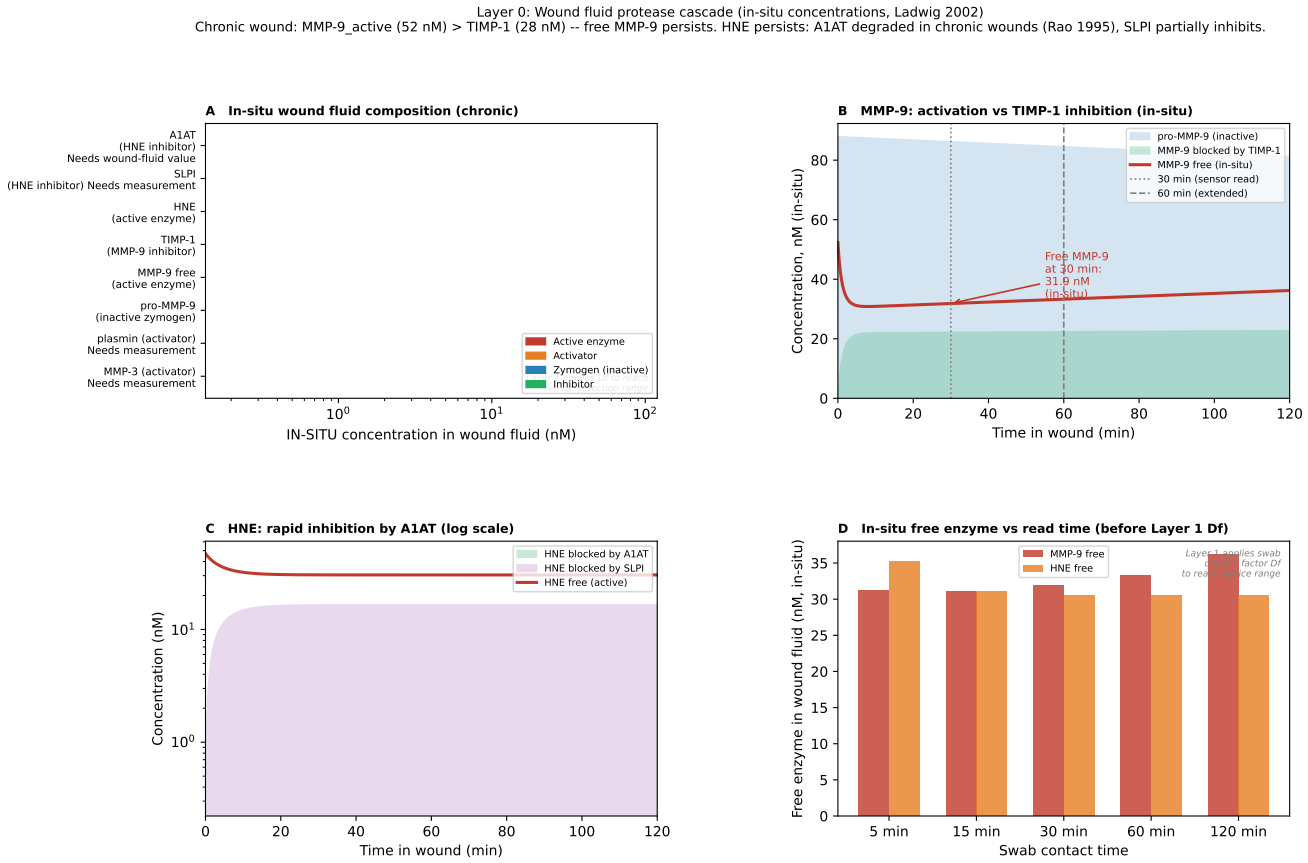


Figure 1: Layer 0 ODE output for the chronic wound state over 120 min. Top left: in-situ initial concentrations. Top right: MMP-9 activation vs. TIMP-1 inhibition. Bottom left: HNE in both wound states (log scale; note 5.5 ms half-life in healing). Bottom right: free enzyme at read times.

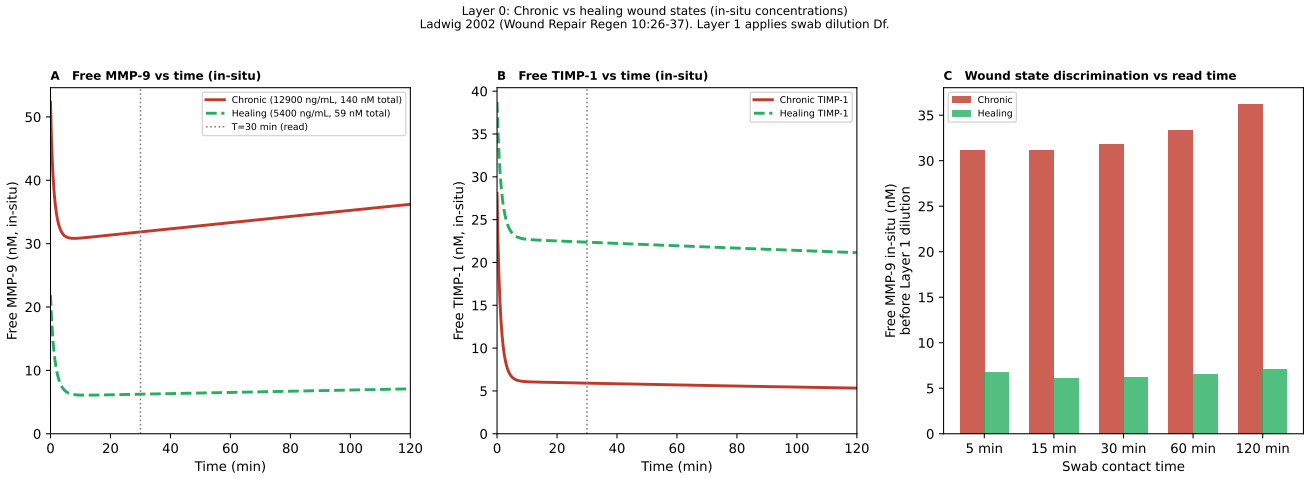


Figure 2: Chronic vs. healing wound states. Left: free MMP-9 over time. Centre: free TIMP-1. Right: MMP-9 discrimination at read times.

ODE results at $t = 30$ min

Species	Chronic wound	Healing wound	C/H ratio
Free MMP-9	31.9 nM	6.2 nM	5.1×
Free TIMP-1	5.9 nM	22.4 nM	–
Free HNE	30.5 nM †	0.00 pM	–
Free A1AT	0.00 μM	1.88 μM	–

Table 2: ODE output at $t = 30$ min. †A1AT degraded in 9/10 chronic samples (Rao 1995 [7]); HNE elevated in only 50% of chronic VLU (Trengove 1999 [1]). Chronic HNE prediction is sensitive to placeholder SLPI kinetics. Healing values depend on unconfirmed initial conditions.

MMP-9 wound-state discrimination

The 5.1× discrimination ratio arises from the sign of the MMP-9 vs. TIMP-1 excess, as made explicit by Eq. (II.9):

- **Chronic wound**: MMP-9 (52 nM) exceeds TIMP-1 (28 nM) by 24 nM. With $K_i = 8.5$ nM $\ll M_0, T_0$, TIMP-1 is nearly fully bound and free MMP-9 persists at ≈ 32 nM.
- **Healing wound** (estimated): TIMP-1 (39 nM) exceeds MMP-9 (22 nM) by 17 nM. TIMP-1 excess suppresses free MMP-9 to ≈ 6 nM.

The mechanism is structurally robust: for $K_i \ll M_0, T_0$, Eq. (II.9) gives $[M]_{\text{free}} \approx M_0 - T_0$ to first order (the excess simply remains free). Any wound state where active MMP-9 exceeds TIMP-1 will show elevated free MMP-9, independent of exact parameter values.

HNE wound-state discrimination

The A1AT wound-state switch (Rao 1995 [7], Section III) produces a strong binary in HNE: healing wounds have $[\text{HNE}]_{\text{free}} = 0.00$ pM, while chronic wounds with degraded A1AT show $[\text{HNE}]_{\text{free}} \approx 30$ nM. However, elevated HNE was observed in only 6/13 (50%) of chronic VLU samples (Trengove 1999 [1]); the remaining 50% of chronic wounds are HNE-low regardless of A1AT status. HNE is not a reliable primary discriminant; MMP-9 (5× discrimination, grounded in confirmed kinetics) is the preferred target. The second electrode targeting HNE serves as an internal reference rather than a primary channel.

Sensor signal implication

The primary output of this model is the free MMP-9 concentration in wound fluid: 30 nM (chronic) and 6 nM (healing) at the 30-min read point. Since sensor signal f_{cleaved} is proportional to $[\text{MMP-9}]_{\text{free}}$ in the linear cleavage regime (Layer 9), the wound-state discrimination ratio is:

$$\frac{[\text{MMP-9}]_{\text{free,chronic}}}{[\text{MMP-9}]_{\text{free,healing}}} \approx \frac{30 \text{ nM}}{6 \text{ nM}} = 5\times \tag{V.1}$$

How $[\text{MMP-9}]_{\text{free}}$ drives surface cleavage of the electrode probe is modelled in Layer 9. The solution-phase kinetic constants for SGKIPRTLTA ($k_{\text{cat}} = 86.4$ s^{-1}, $K_m = 1.28$ mM, Kridel 2001 [9]) are confirmed inputs to that layer.

Layer 1 dilution

The in-situ free MMP-9 concentrations from Table 2 (30 nM chronic, 6 nM healing) are the direct inputs to the Layer 1 dilution calculation, which determines what sample dilution factor is needed for the device to operate within its detection window. The required dilution factor and LOD are specified in Layer 1 once the surface kinetics (Layer 9) are characterised.

Limitations of this model

Each limitation below states what is simplified, explains why the simplification does not invalidate the model's primary output (the MMP-9 wound-state discrimination ratio), and identifies what experiment would resolve it.

1. **Healing wound initial conditions are not text-confirmed.** The chronic wound values are the primary output of this model: active MMP-9 = 52 nM and TIMP-1 = 28 nM are both text-confirmed from Ladwig 2002 (p. 31) [2], and all kinetic parameters (K_i, k_{on}^T, k_{off}^T) are text-confirmed from Olson 1997 Table IV [4]. The healing MMP-9 estimate (5,400 ng/mL) is read from Ladwig 2002 Figure 8; the healing TIMP-1 (1,104 ng/mL) has no paper source. These estimates are used only to compute a comparison point, not to determine the chronic wound output. The model's structural conclusion, that any wound state in which active MMP-9 exceeds TIMP-1 will yield elevated free MMP-9 (from the sign of Eq. (II.9)), is independent of the healing estimates. The 5× ratio should be treated as a working figure. Resolution requires a joint VLU dataset reporting both MMP-9 and TIMP-1 from the same patients at chronic and healing time points.

2. **SLPI kinetics are unconfirmed placeholders and do not affect MMP-9.** The SLPI sub-system (Eqs. IV.8–IV.9) is completely decoupled from the MMP-9/TIMP-1 sub-system: SLPI does not bind MMP-9 or TIMP-1 and cannot alter the MMP-9 partition calculated in Section II. SLPI uncertainty propagates exclusively to the HNE secondary channel, which is already acknowledged as a limited discriminant (Limitation 4 below). The SLPI concentration and kinetics ($k_{\text{on}}^S = 10^5$ M^{-1}s^{-1}, $k_{\text{off}}^S = 10^{-4}$ s^{-1}) are order-of-magnitude placeholders; Ashcroft 2000 [6] confirms SLPI's role in wound healing from knockout mice but provides no quantitative wound-fluid SLPI concentration. Wound-fluid SLPI ELISA would resolve this for the HNE channel.

3. **Pro-MMP-9 activation is negligible at current parameter values.** The active MMP-9 initial condition (52 nM, chronic; 22 nM, healing) is taken directly from Ladwig 2002 [2] and does not depend on the activation sub-model. The activation kinetics (via MMP-3 and plasmin; Eq. IV.1) use order-of-magnitude placeholders, and with these values less than 1% of pro-MMP-9 converts in 120 min. This means the active MMP-9 pool is effectively constant over the 30-minute read window, consistent with using the Ladwig ELISA snapshot as the initial condition. These activation parameters become important only if wound-fluid MMP-3 or plasmin concentrations prove substantially higher than assumed, which would require a separate measurement.

4. **HNE elevation is inconsistent across chronic wound samples.** Elevated free HNE was detected in only 6 of 13 chronic VLU samples in Trengove 1999 [1]: 50% of chronic wounds are HNE-low regardless of A1AT status. The HNE channel therefore has limited diagnostic utility and is not the primary discriminant. This limitation does not affect MMP-9, which shows elevated free concentration in all chronic wound samples by the TIMP-1/MMP-9 excess mechanism (Section II). The HNE electrode is treated as a secondary reference channel, not a primary diagnostic output.

5. **Solution-phase kinetics for SGKIPRTLTA are confirmed from Kridel et al. (2001) [9]; surface immobilisation effects are addressed in Layer 9.**

   Kridel et al. (2001) [9] Table II reports solution-phase kinetics for MMP-9 cleaving peptide A11 (SGKIPRTLTA):

   $$k_{\text{cat}} = 86.4 \text{ s}^{-1}, \quad K_m = 1.28 \text{ mM}, \quad k_{\text{cat}}/K_m = 67{,}500 \text{ M}^{-1}\text{s}^{-1}$$

   measured by fluorescamine incorporation assay at pH 7.5, 10 mM $CaCl_2$, 37°C. These values confirm that SGKIPRTLTA is an efficient MMP-9 substrate in solution. How surface immobilisation modifies these kinetics is the subject of Layer 9; the solution-phase values serve as an upper bound for the surface effective rate constant.

   For HNE: A1AT inhibition is a serpin acyl-enzyme mechanism, not competitive inhibition. The half-life conclusion is insensitive to a 10-fold change in k_{on}^A (Eq. III.6). The k_{cat} and K_m for HNE cleaving the HNE peptide substrate are not yet confirmed from any paper in `papers/layer_00/` and remain an outstanding parameter for the HNE channel.

References

All papers in `electrochemical_device/papers/layer_00/`.

[1] Trengove NJ, Stacey MC, Macauley S, Bennett N, Gibson J, Burslem F, Murphy G, Schultz G (1999). Analysis of the acute and chronic wound environments: the role of proteases and their inhibitors. *Wound Repair Regen* 7(6):442–452. **trengove1999.pdf** – TIMP-1 and HNE values from text (p. 448) confirmed.

[2] Ladwig GP, Robson MC, Liu R, Kuhn MA, Muir DF, Schultz GS (2002). Ratios of activated matrix metalloproteinase-9 to tissue inhibitor of metalloproteinase-1 in wound fluids are inversely correlated with healing of pressure ulcers. *Wound Repair Regen* 10(1):26–37. **ladwig2002.pdf** – chronic MMP-9 and TIMP-1 from text (p. 31) confirmed; healing values from Figure 8 only (approximate).

[3] Nagase H, Woessner JF Jr (1999). Matrix metalloproteinases. *J Biol Chem* 274(31):21491–21494. **nagase_woessner1999.pdf** – biological context; no kinetic constants used.

[4] Olson MW, Gervasi DC, Mobashery S, Fridman R (1997). Kinetic analysis of the binding of human matrix metalloproteinase-2 and -9 to tissue inhibitor of metalloproteinase (TIMP)-1 and TIMP-2. *J Biol Chem* 272(47):29975–29983. **olson1997.pdf** – k_{on}^T, k_{off}^T, K_i for MMP-9/TIMP-1 confirmed from Table IV.

[5] Beatty K, Bieth J, Travis J (1980). Kinetics of association of serine proteinases with native and oxidized α-1-proteinase inhibitor and α-1-antichymotrypsin. *J Biol Chem* 255(9):3931–3934. **beatty1980.pdf** – $k_{\text{on}}^A = 6.5 \times 10^7$ M^{-1}s^{-1} confirmed from Table 1.

[6] Ashcroft GS, Lei K, Jin W, Longenecker G, Kulkarni AB, Greenwell-Wild T, Hale-Donze H, McGrady G, Song XY, Wahl SM (2000). Secretory leukocyte protease inhibitor mediates non-redundant functions necessary for normal wound healing. *Nat Med* 6(10):1147–1153. **ashcroft2000.pdf** – SLPI role confirmed in knockout mouse model; no quantitative wound-fluid SLPI concentration reported.

[7] Rao CN, Ladin DA, Liu YY, Chilukuri K, Dong Z, Bhagwat SV (1995). α_1-Antitrypsin is degraded and non-functional in chronic wounds but intact and functional in acute wounds: the inhibitor protects fibronectin

from degradation by chronic wound fluid enzymes. *J Invest Dermatol* 105(4):572–578. **rao1995.pdf** – A1AT degraded in 9/10 chronic samples, intact in healing, confirmed from text (pp. 572–575). No quantitative wound-fluid A1AT concentration given.

[8] Nwomeh BC, Liang HX, Cohen IK, Yager DR (1999). MMP-8 is the predominant collagenase in healing wounds and nonhealing ulcers. *J Surg Res* 81(2):189–195. **nwomeh1999.pdf** – TIMP-1 "very low or not detected" in chronic ulcer fluid confirmed from text (p. 193). Units not convertible to ng/mL without protein content.

[9] Kridel SJ, Chen E, Kotra LP, Howard EW, Mobashery S, Smith JW (2001). Substrate hydrolysis by matrix metalloproteinase-9. *J Biol Chem* 276(23):20572–20578. DOI: 10.1074/jbc.M100900200. **kridel2001.pdf** – $k_{\text{cat}} = 86.4\ \text{s}^{-1}$, $K_m = 1.28$ mM, $k_{\text{cat}}/K_m = 67{,}500\ \text{M}^{-1}\text{s}^{-1}$ for peptide A11 (SGKIPRTLTA) by MMP-9 confirmed from Table II. Measured by fluorescamine assay, pH 7.5, 10 mM $CaCl_2$, 37°C.